

Stochastic Modelling of Traffic Accident Risk Using Markov Chains

Accident risk modelling has become increasingly important in improving public safety and supporting decision-making in areas such as transportation, environmental management, and public policy. By analysing how accident risks change over time and identifying factors that influence accident severity, researchers and policymakers are able to develop more effective prevention strategies and safety interventions. In addition, accident risk modelling helps support resource allocation and policy planning and reduces the likelihood of severe accidents.

One of the most common approaches used is the generalized nonlinear model (GNM). This model applies a nonlinear regression function to better capture non-monotonic relationships between the independent and dependent variables. Lao et al. (2014) used road accident data collected from 10 highway routes from 2002 through 2006 to examine this approach. The results showed that GNM offers more flexibility compared to GLM to uncover the nonlinear relationship within the data, and therefore offers a more reasonable explanation of accident risk.

Another approach that is suitable for modelling risk is a Markov chain. In fields such as engineering and economics, the usage of Markov chains as a prediction technique is already common, especially because it allows simulations and supports more informed decision-making. According to research (Moreno-Sanf  lix et al., 2024), this model makes it possible to analyse stochastic processes that change randomly over time and move across a range of possible states without memory of past states.

However, in scientific literature, there are limited works on traffic accident prediction using Markov chains. Therefore, this assignment is aimed at demonstrating how Markov chains can be effectively used in fields other than engineering and economics, mainly focused on accident risk. Some relevant literature in the road crash field is shown below.

Lin et al. (2011) analysed 5,354 multi-year observations covering twelve socio-economic and built environment factors to predict pedestrian road traffic accidents using Markov chain and cellular automata Markov chain models. Their findings showed that transition-based models were capable of capturing changes in accident risk over time and identifying persistent high-risk conditions. Udo & Etika (2025) applied a Markov chain model to analyse traffic law enforcement and accident rates, demonstrating the usefulness of transition-based modelling in road safety analysis.

To further understand the application of Markov chains in accident risk modelling, it is important to understand the underlying theory and properties of Markov chains. According to the University of Auckland (n.d.) and the University of Chicago (n.d.) Markov chains satisfied 2 definitions or conditions

The first condition is the existence of a transition probability kernel, also referred to as a probability matrix or a stochastic matrix, on a finite or countable set of X is a matrix $\mathbb{P}=(p(x,y))_{x,y \in X}$, where

each entry is a non-negative value. The rows and columns are indexed by the element of X , and the entries in each row must sum to 1, such as for $x \in X$

$$\sum_{y \in X} p(x, y) = 1$$

The second condition is a discrete-time Markov chain with a finite or countable state space X and transition probability matrix $\mathbb{P}=(p(x,y))_{x,y \in X}$ is a sequence of random variables $X_0, X_1, X_2, \dots, X_n$ that take any values in X , where the probability of the next state depends only on the current state. Or mathematically, for any $x \in X$ and any integer $n = 0, 1, 2, \dots, n$

$$\mathbb{P}(X_{n+1} = y \mid X_n = x_n, \dots, X_0 = x_0) = \mathbb{P}(X_{n+1} = y \mid X_n = x_n)$$

This follows directly from the definition of conditional probability, where

$$P(A|B) = \frac{p(A \cap B)}{P(B)}$$

With condition $P(B) > 0$. This shows that once the present state is known, the earlier history does not affect the next step. As a result, the probability of a sequence of states can be found by multiplying the initial probability by the relevant transition probabilities, such that

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mathbb{P}(X_0 = x_0) \prod_{i=1}^n p(x_{i-1}, x_i)$$

This follows because, by the Markov property,

$$\mathbb{P}(X_i = x_i \mid X_0 = x_0, \dots, X_{i-1} = x_{i-1}) = p(x_{i-1}, x_i)$$

In general, a discrete-time Markov chain can be described by the transition matrix

$$\mathbb{P}_{i,j}^{m,n} = \mathbb{P}(X_n = j \mid X_m = i)$$

For $n > m$. The one-step case is

$$\mathbb{P}_{i,j}^{m,n} = \mathbb{P}(X_{m+1} = j \mid X_m = i)$$

If the one-step transition probability is independent of the time index m , then the process is called a time-homogeneous Markov chain. In case,

$$\mathbb{P}(X_{m+1} = j \mid X_m = i) = p_{ij}$$

And the transition probabilities remain constant over time.

In this assignment, a homogeneous discrete-time Markov chain framework is adopted because the monthly transition probabilities are assumed to remain constant throughout the observation period from January 2018 to March 2026. This assumption simplifies the modelling process and allows a single transition matrix to describe the system dynamics. The model is applied to two variables: monthly road fatality risk states and rainfall-based weather regimes.

To construct the Markov chain, the continuous observations are converted into discrete states. This allows the transitions between levels of accident risk and weather conditions to be analysed over time.

For the monthly road fatality data, the states are categorised using the 33rd and 66th percentiles of the observations, which are 97 and 106.92, respectively. Hence, the fatality risk states are defined as

$$X_t = \begin{cases} LowFR, & \text{if } fatalities_t < 97 \\ MediumFR, & \text{if } 97 \leq fatalities_t \leq 106.92 \\ HighFR, & \text{if } fatalities_t > 106.92 \end{cases}$$

Similarly, the rainfall states are categorised using the 33rd and 66th percentiles of the rainfall data, which are 37.6 and 63.528 respectively:

$$Y_t = \begin{cases} LowWC, & \text{if } rainfalls_t < 37.6 \\ MediumWC, & \text{if } 37.6 \leq rainfalls_t \leq 63.528 \\ HighWC, & \text{if } rainfalls_t > 63.528 \end{cases}$$

The transitional probabilities are estimated empirically from the observed data using

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_j n_{ij}}$$

Where n_{ij} denotes the number of observed transitions from state I to state j and $\sum_j n_{ij}$ represents the total number of transitions originating from state i.

Thus, the transition matrix is written as

$$P = \begin{pmatrix} PLL & PLM & PLH \\ PML & PMM & PMH \\ PHL & PHM & PHH \end{pmatrix}$$

Where each row sums to 1 because the entries represent all possible future states conditional on the current state.

Using the R code provided, the transition probability matrix for the monthly road fatality risk states is obtained as follows :

$$P_{FR} = \begin{array}{ccccc} & & Low & Medium & High \\ Low & 0.3333333 & 0.3333333 & 0.3333333 & \\ Medium & 0.4375000 & 0.2500000 & 0.3125000 & \\ High & 0.3636364 & 0.2272727 & 0.4090909 & \end{array}$$

Similarly, the transition probability matrix for the rainfall weather regimes is

$$P_{WC} = \begin{array}{ccccc} & & Low & Medium & High \\ Low & 0.3684211 & 0.3421053 & 0.2894737 & \\ Medium & 0.3548387 & 0.3870968 & 0.2580645 & \\ High & 0.2954545 & 0.1590909 & 0.5454545 & \end{array}$$

The transition matrices provide insight into how likely the process is to remain in the same state or move to another state over time. One interesting finding is that the transition probabilities from the “LowFR” state are exactly equal. Specifically, if the process is currently in the low fatality risk state, there is an equal probability of 0.3333 moving to the low, medium, or high state in the following month. This suggests that the low-risk state does not display strong persistence and behaves relatively unpredictably compared to the other states. Such a pattern may suggest that periods of low accident risk are relatively unstable and more easily change depending on external conditions such as traffic volume, seasonal effects, or weather changes.

For the “MediumFR” state, the largest probability is transitioning to the “LowFR” state with probability 0.43750, followed by transitioning to “HighFR” with probability 0.3125, while the probability of remaining in the medium state is only 0.2500. This suggests that the medium-risk state is relatively temporary and tends to shift toward either lower or higher risk conditions rather than remaining stable over time.

For the “HighFR” state, the probability of remaining in the same state is 0.4091, which is the largest probability in that row. This indicates persistence in high accident risk periods, meaning that once road fatality risk becomes high, it is relatively likely to remain high in the following month. This finding is significant from a policy perspective because persistent high-risk periods may require targeted interventions or preventive measures to reduce continued accident severity.

On the other hand, unlike the fatality risk matrix, the rainfall transition matrix displays stronger diagonal dominance, particularly in the “HighWC” state. The probability of remaining in the high rainfall state is 0.5455, which is substantially larger than the probabilities of transitioning to low rainfall (0.2955) or medium rainfall (0.1591). This suggests that periods of heavy rainfall tend to persist across consecutive months, indicating relatively stable weather regimes once high rainfall conditions occur.

The “MediumWC” state also demonstrates moderate persistence, with a probability of 0.3871 of remaining in the same state. Meanwhile, the “LowWC” state has a probability of 0.3684 of remaining low, which is only slightly larger than the probability of moving to the medium state (0.3421). This suggests that lower rainfall conditions are less stable and can transition relatively easily into moderate rainfall conditions.

Overall, the rainfall process appears more persistent and stable than the road fatality risk process, particularly in the high state. In contrast, the fatality risk process demonstrates more variability, especially within the low and medium states. These findings are interesting because they suggest that weather regimes may follow more stable transition dynamics, whereas road fatality risk may be influenced by a wider range of unpredictable behavioural, environmental, or traffic-related factors.

To be able to see if there is any dependence between those processes, we need to construct a joint matrix of fatality risk and weather conditions at each time period. Since we only have 3 states for each model, the total will be 9 joint states, resulting in a 9x9 matrix, corresponding to all possible combinations of Low, Medium, and High states.

The transition probabilities for the joint Markov chains are defined as

$$\mathbb{P}_{(i,j),(k,l)} = \mathbb{P}(X_{t+1} = k, Y_{t+1} = l \mid X_t = i, Y_t = j)$$

Where X_t represents the road fatality risk state at time t , and Y_t represents the weather condition state at time t . Here (i, j) represents the current joint state, and (k, l) represents the next joint state.

A transition matrix was then formed by counting the movements from each joint state at time t to the joint state at time $t+1$, and converting these counts into probabilities by dividing each row by its total. Mathematically:

$$\hat{p}_{(i,j),(k,l)} = \frac{N_{(i,j),(k,l)}}{\sum_{(a,b)} N_{(i,j),(a,b)}}$$

Where $N_{(i,j),(k,l)}$ denotes the number of observed transitions from state (i,j) to state (k,l) .

Using the R code provided, the estimated transition matrix is obtained as shown below:

	HighRF_HighWC	HighRF_LowWC	HighRF_MediumWC	LowRF_HighWC	LowRF_LowWC	LowRF_MediumWC
HighRF_HighWC	0.23076923	0.15384615	0.07692308	0.15384615	0.23076923	0.00000000
HighRF_LowWC	0.09090909	0.00000000	0.27272727	0.18181818	0.09090909	0.09090909
HighRF_MediumWC	0.08333333	0.16666667	0.08333333	0.08333333	0.16666667	0.16666667
LowRF_HighWC	0.26315789	0.10526316	0.00000000	0.21052632	0.05263158	0.00000000
LowRF_LowWC	0.12500000	0.06250000	0.06250000	0.00000000	0.31250000	0.12500000
LowRF_MediumWC	0.00000000	0.00000000	0.27272727	0.18181818	0.00000000	0.18181818
MediumRF_HighWC	0.00000000	0.25000000	0.16666667	0.33333333	0.00000000	0.08333333
MediumRF_LowWC	0.00000000	0.09090909	0.09090909	0.18181818	0.18181818	0.09090909
MediumRF_MediumWC	0.12500000	0.00000000	0.12500000	0.12500000	0.25000000	0.25000000
	MediumRF_HighWC	MediumRF_LowWC	MediumRF_MediumWC			
HighRF_HighWC	0.15384615	0.00000000	0.00000000			
HighRF_LowWC	0.09090909	0.09090909	0.09090909			
HighRF_MediumWC	0.08333333	0.08333333	0.08333333			
LowRF_HighWC	0.21052632	0.05263158	0.10526316			
LowRF_LowWC	0.12500000	0.12500000	0.06250000			
LowRF_MediumWC	0.09090909	0.27272727	0.00000000			
MediumRF_HighWC	0.00000000	0.08333333	0.08333333			
MediumRF_LowWC	0.09090909	0.09090909	0.18181818			
MediumRF_MediumWC	0.00000000	0.12500000	0.00000000			

The dependence between the two processes can be seen from the transition probabilities. If the two processes were independent, the joint transition probabilities would factorise into the product of the marginal transition probabilities. However, the observed joint transitions do not appear to satisfy this property, suggesting meaningful dependence between weather conditions and fatality risk.

For example, under independence, the probability of transitioning from the HighFR_HighWC state to itself should approximately equal the product of the corresponding marginal probabilities:

$$P(\text{HighFR} \rightarrow \text{HighFR}) \times P(\text{HighWC} \rightarrow \text{HighWC})$$

Using the estimated marginal matrices :

$$0.4091 \times 0.5455 = 0.2231$$

However, the estimated joint transition probability is approximately 0.2381, which differs from the factorised value. This suggests that the independence assumption may not hold exactly and provides evidence of dependence between weather conditions and road fatality risk.

The probability of the next state is not the same for every current state, but changes depending on the current combination of road fatality risk and weather conditions (joint state). For example, if there is high RF (road fatality) and high WC (Weather condition), it has a relatively large probability(0.238) of remaining in high-risk states, which suggests persistence in dangerous conditions. Similarly, if Low Low (low RF and low WC) has a relatively large probability (0.1818) of remaining in low-risk states, this suggests stability in safer conditions.

Therefore, the joint Markov chain indicates dependence because the future movement of road fatality risk and weather conditions depends on the current joint state, rather than evolving independently. This suggests that weather conditions may influence fatality risk through factors such as reduced visibility, slippery roads, or hazardous driving conditions. However, it is less realistic to interpret the relationship in the opposite direction, since fatality risk itself does not influence weather conditions.

The transition matrix assumes that the transition probabilities are constant over the full sample period. However, these parameters may change over time due to seasonal weather patterns, changes in road safety policy, traffic behaviour, or infrastructure improvements. Therefore, the estimated matrix should be interpreted as an average description of the transition dynamics rather than fixed probabilities that necessarily remain unchanged in every period. This suggests the process may not be strictly stationary over the full sample period, meaning the homogeneous Markov assumption may only hold approximately

Determining which model is better, either the independent models (Fatality + Weather) or the integrated joint model (Fatality, Weather), needs to be done by examining and considering several points.

On one hand, Hempel et al. (2021) show that an independent model offers simplicity and is easier to interpret. They may not fully capture the dependence between the processes. Independent

models assume that weather conditions and fatality risk evolve separately, which may not hold in reality because adverse weather conditions, such as heavy rain or storms, may increase the likelihood of higher road fatality risk. The main advantages of independent models are their lower computational burden, simpler estimation process, and easier interpretation. In addition, separating the process reduces the dimensionality of the state space and avoids the combinatorial explosion problem that often arises in joint models. However, an important limitation is that independent models ignore interdependent effects between weather and fatality regimes, which may lead to biased or less accurate transition probability estimates if the variables are strongly dependent.

On the other hand, the integrated joint model combines fatality risk and weather conditions into a single process, allowing the dependence structure between the two variables to be captured directly. According to research from Rustand et al. (2023), joint models are useful when variables are correlated because they reduce bias in parameter estimation and improve efficiency by utilizing all available information simultaneously. However, joint models also have limitations, including higher computation cost, increased model complexity, and separate transition problems where some state transitions occur very rarely, causing unstable probability estimates.

In practice, the joint model may provide better predictive performance when weather and fatality risk are strongly associated, since dependence information is retained. However, when dependence is weak or sample sizes are limited, independent models may perform more reliably due to more stable transition probability estimates because each process is estimated separately using a smaller state space. In contrast, joint models often suffer from sparse transition counts, especially when certain state combinations occur infrequently. This may reduce estimation reliability despite the joint framework having captured more dependence structure.

In this study, the joint model allows combinations such as High Fatality, Storm, or Low Fatality, Clear Weather to be analysed explicitly through a transition matrix. Since this study only involves two variables with three states each, resulting in a manageable 9x9 transition matrix, the computational complexity remains relatively manageable. Therefore, the joint model is preferable because the dependence between weather and fatality risk is likely significant.

According to the PTC Help Centre (2024), there are several limitations of using a Markov chain. First of all, it only takes into account the current state, but ignores the historical data. Meanwhile, in reality, road fatality risk may be influenced by a prolonged or lag factor that the Markov model didn't take into account, or in other words, the Markov model oversimplified. Furthermore, Markov's model is most appropriate when transition probabilities can reasonably be assumed constant over time. In reality, the relationship may change over time, causing one to assume it is constant is too unrealistic.

In addition, Del Sole (2000) states that another limitation of the Markov model is that it may be inappropriate when the time interval is too short. In such cases, changes between states are highly

deterministic rather than random, meaning that the system evolves smoothly over time. As a result, the probabilistic transition structure assumed in a Markov chain may not accurately represent the true dynamics.

According to Sharma (2026), another alternative modelling framework is the Hidden Markov Model (HMM), which extends the standard Markov chain by allowing the underlying states to remain unobserved (hidden) while generating observable outcomes. In this context, the hidden states may represent the true underlying road risk levels, which are inferred from observable variables such as weather conditions and road fatality data. Unlike the standard Markov chain, HMM allows uncertainty in state classification, which provides a more realistic representation of latent accident risk dynamics.

Another alternative is to use a non-homogeneous or time-varying Markov chain, where transition probabilities are allowed to change over time, making the model more flexible in capturing seasonal or structural changes. Logistic regression or multinomial logistic regression may also be applied to estimate the probability of entering a high fatality regime based on weather conditions and other explanatory variables. For example, the hypothesis

$$H_0: \beta_{WEATHER} = 0$$

$$H_1: \beta_{WEATHER} \neq 0$$

To test whether weather conditions significantly affect fatality risk. If the null hypothesis is rejected, it would suggest that weather conditions have a statistically significant influence on road fatality regimes. Time series approaches, such as ARIMA, may also be useful because they incorporate lagged effects, allowing past weather conditions to influence future fatality risk.

Overall, these models may provide valuable insight for real-world decision-making and risk forecasting. By identifying persistent high-trend weather regimes, authorities may implement targeted interventions such as temporary speed restrictions, increased road patrols, or safety warnings during dangerous weather conditions. Furthermore, understanding the dependence between weather and fatality risk may improve forecasting accuracy and support resource allocation for emergency response and traffic management.

Use of Generative AI

Grammarly and ChatGPT were used in a limited capacity to support proofreading, paraphrasing, and checking alignment with the assignment requirements and marking rubric. The final work, including all analysis, interpretation, and conclusions, represents the author's own understanding and original work.

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